



Technical Service Bulletin

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Temporary KV Rear Crankshaft Seal Tool

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Warranty Statement

The information in this document has no effect on present warranty coverage or repair practices, nor does it authorize TRP or Campaign actions.

Contents

Recently a new rear crankshaft seal was released for use on K38/50 and QSK38/50 liter engines. The new seal required new tooling for installation and removal. The new tooling was unavailable at parts distribution centers when the seal was introduced. The following document instructs how to modify the current QSK45/60 crankshaft seal tool, Part Number 3163349, to enable it to install the new KV rear crankshaft seal.

Note : All the dimensions and measurements in this document are in millimeters.

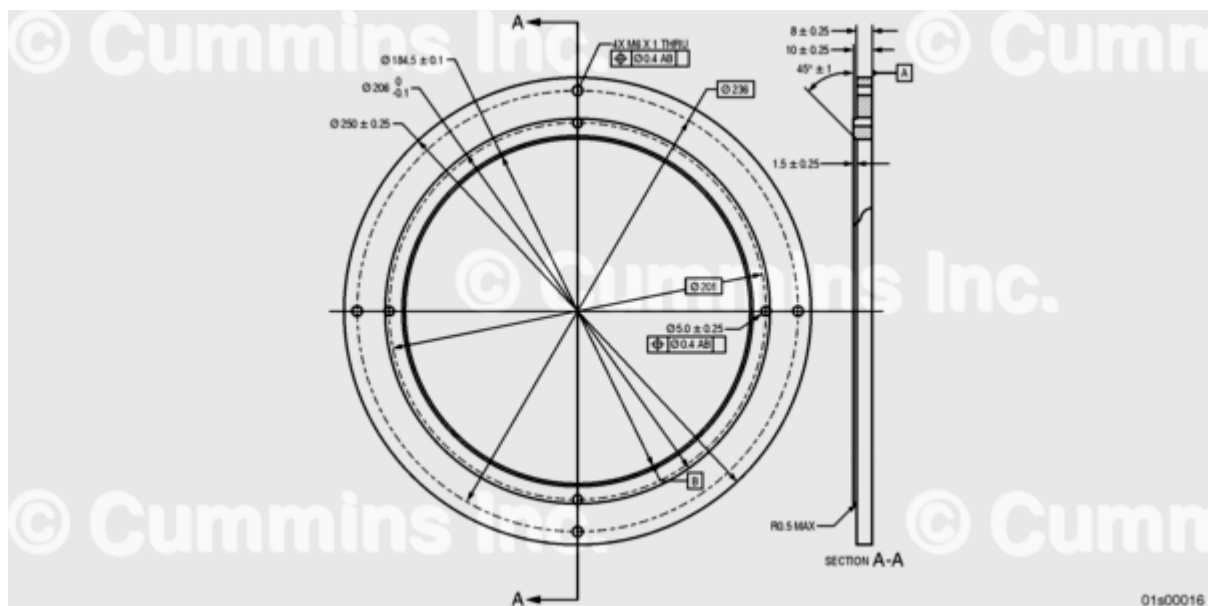


Figure 1, New adaptor plate (dimensions in mm)

Step 1 - Fabricate an adaptor plate as per the above drawing.

Note : This is a short term solution until the new tool is readily available at parts distribution centers. The dimensions in the drawing below are in mm.

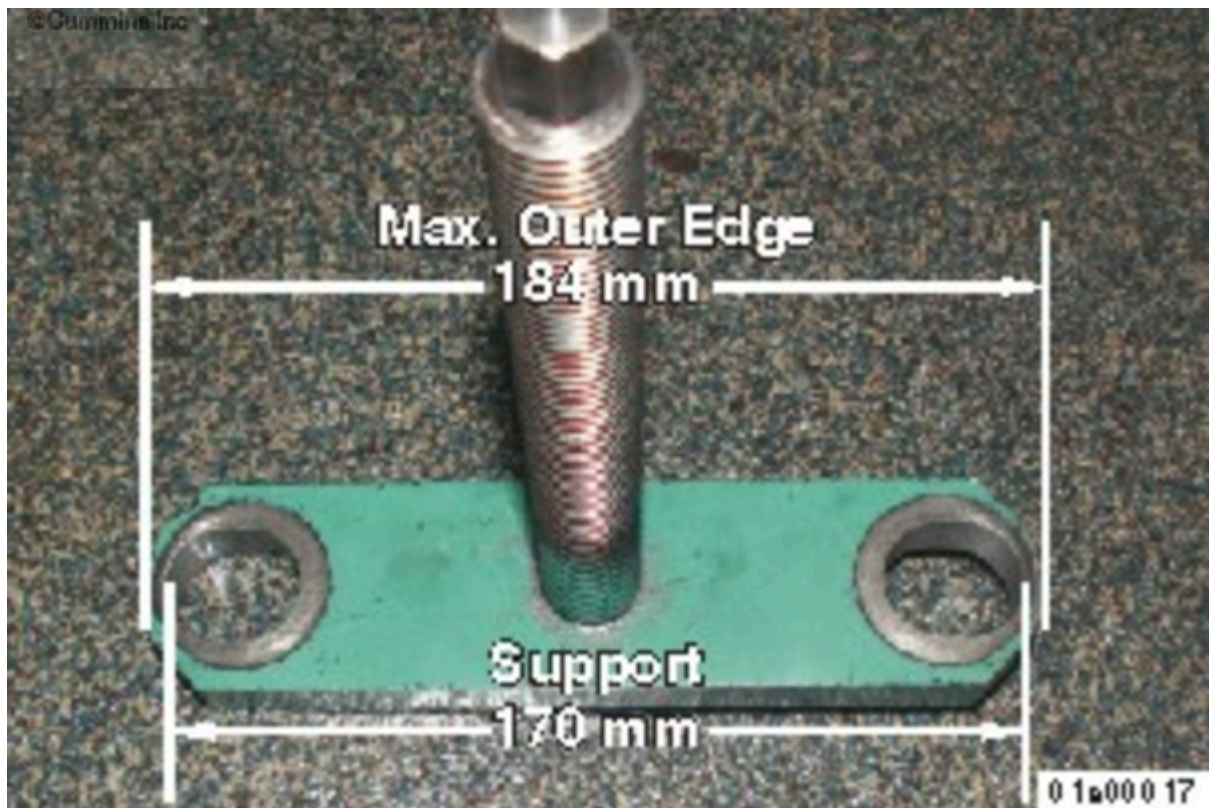


Figure 2, QSK45/60 service tool, support modification

Step 2 - Modify the current QSK45/60 support and spacers.

- Elongate the support mounting holes (outwards) to the dimension shown
- File the outer edges off the support; the outside diameter **must not** exceed 184 mm
- Reduce the spacer height to 25 mm.

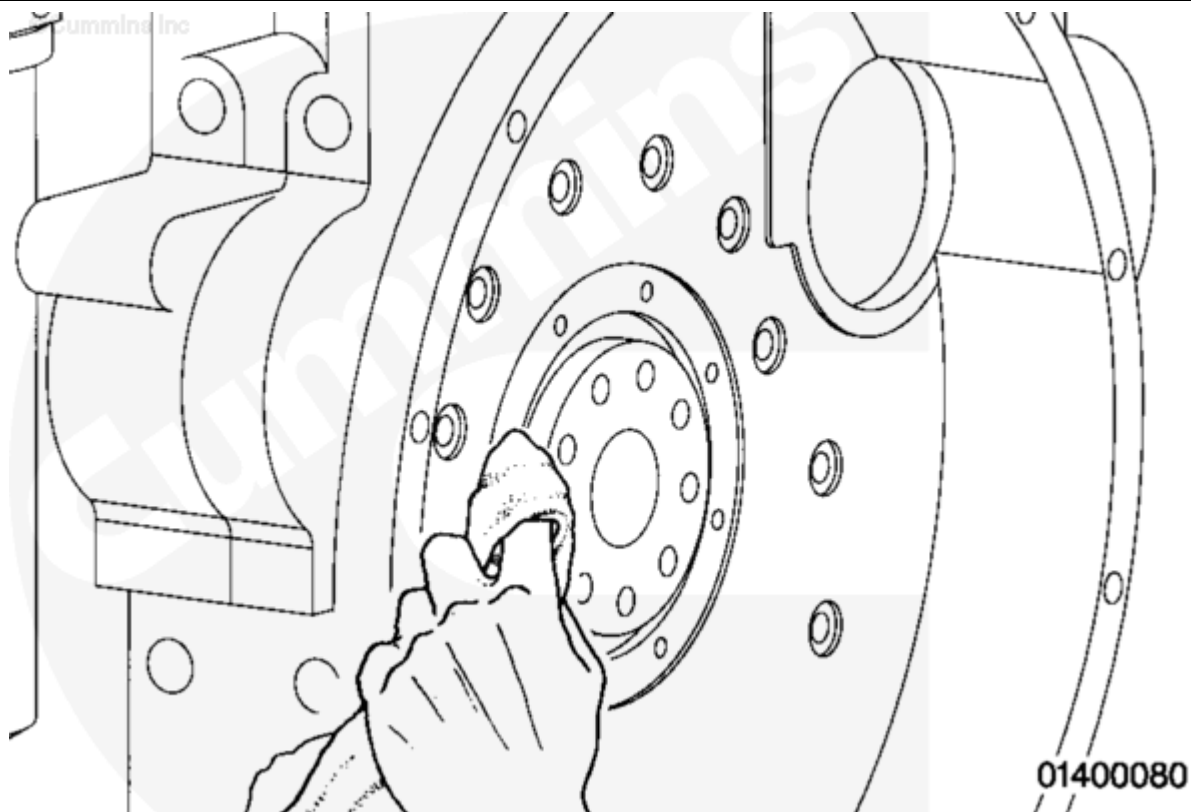
Step 3 - Mount adaptor plate to QSK45/60 driver unit.

- The stepped edge faces away from the tool
- Use 4 of M6x1x14 mm capscrews (Part Number 3900627)
- Align the outside diameter of the adaptor plate with the outside diameter of the driver unit.



Figure 3, Installation of adaptor plate to QSK45/60 driver unit.

Step 4 - Seal Installation.



Make certain the crankshaft and flywheel housing seal mounting areas are clean.



Place the seal on the crankshaft and lightly tap with a soft mallet to start the seal in place.

Note : The seal part number is printed on the front face of the seal. The part number must face away from the cylinder block when installed.

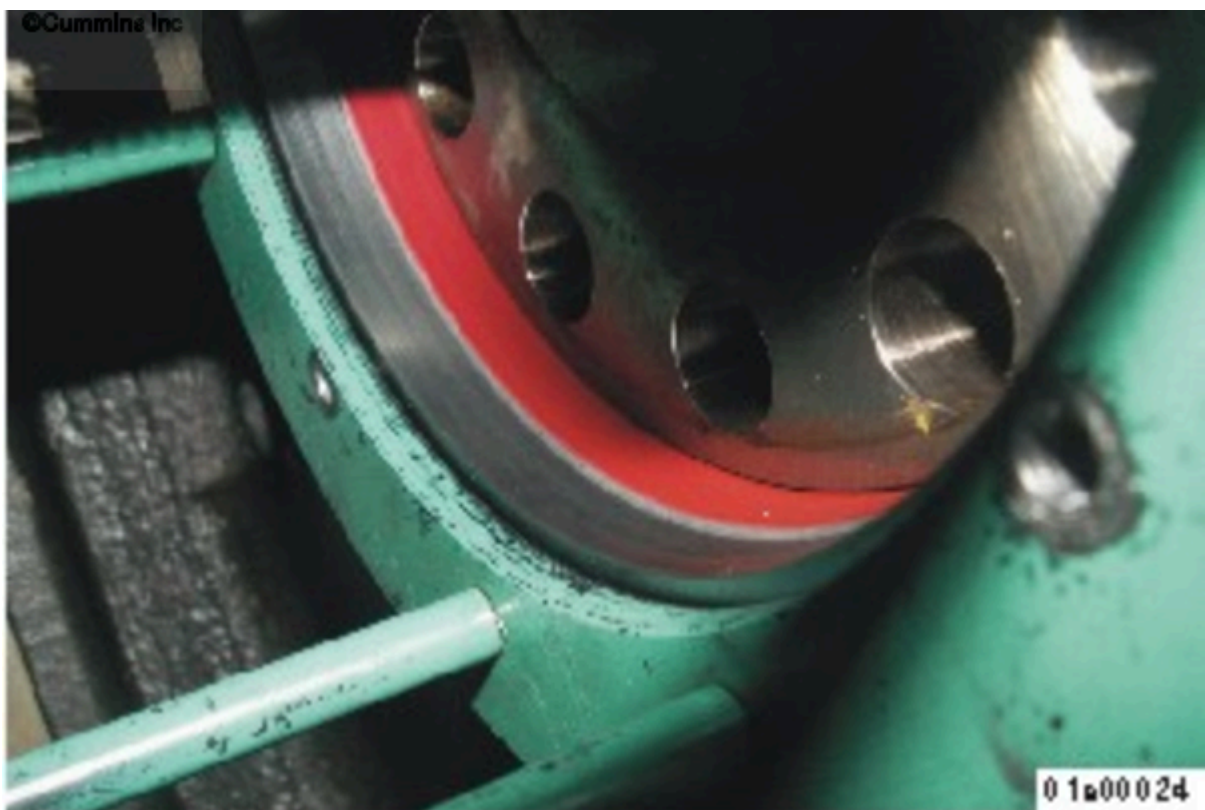


Install the tool onto the crankshaft.

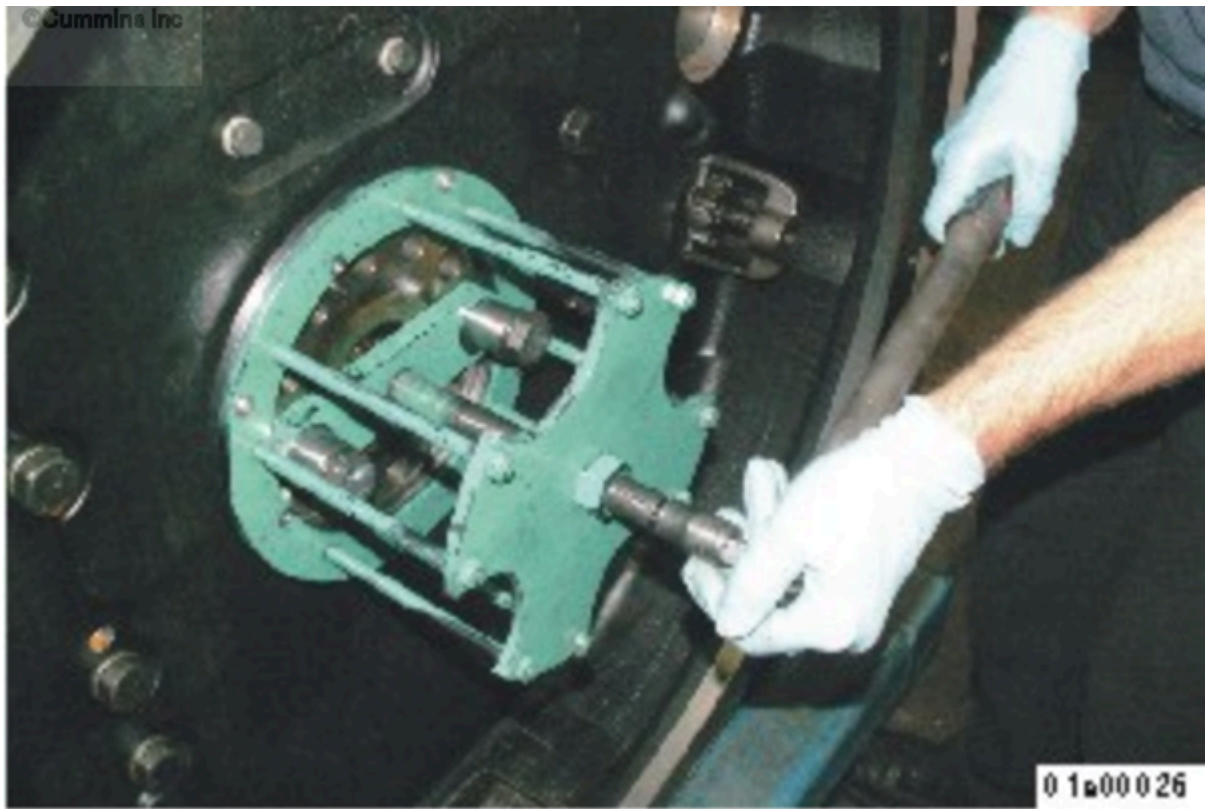
Make certain the support is centered on the crankshaft and clear of the seal.



Install the driver unit on the support until it contacts the seal.



Align the driver unit with the seal.



Start to press the seal onto the crankshaft.



Note : As the seal is pressed on, the driver can move slightly out of alignment with the seal. Make certain to realign the driver unit as it gets close to the crankshaft.

A small chamfer was added to the adapter to help guide the driver unit onto the crankshaft.



Continue to press the seal into place until the adapter unit contacts the flywheel housing.

Approximate torque: 60 Nm [44 ft.lb.]

Make certain the seal driver unit is in full contact with the flywheel housing.



Remove the driver unit and support from the crankshaft.

The seal will be pressed approximately 2 mm below the face of the flywheel housing.

Document History

Date	Details
2010-11-11	Module Created

Last Modified: 30-Nov-2010
